Model-Based Clustering of Climatological Functional Data with Temporal Dependence

An Autoregressive Mixture Model for Bangladesh Monsoon Rainfall

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Abstract

A climate record can be treated as a set of curves. Each grid cell gives one within-season rainfall curve per year, and a mixture model can group the cells by the shape of those curves. The usual versions assume that the annual curves of one cell are independent once its cluster is known. Rain in one year often resembles rain in the next, so that assumption is doubtful. A small pilot (Section 5.1) shows one cost: when the dependence is ignored, the Bayesian information criterion (BIC) tends to pick too many clusters. The pilot also shows a case where modelling the dependence does worse, and the full study is built to find out when that happens.

I propose a likelihood-based mixture model for repeated annual curves. Each spatial unit keeps one cluster label for the whole study period. Within a unit, the principal component scores of the curves follow a Gaussian AR(1) process. All clusters share one autoregressive coefficient, while the means and innovation covariances differ by cluster. I estimate the model by EM, using a Prais–Winsten transformation whose Jacobian stays in the likelihood.

A simulation study will compare the model with an independence-based mixture and with a mixture that allows persistent unit-level offsets, while dependence, cluster separation, imbalance, heavy tails and variance change vary. I will then fit the model to ERA5-Land daily precipitation for Bangladesh, 1950–2025. The project does not claim a new model class. It asks when, and by how much, modelling year-to-year dependence improves functional clustering.

Keywords: functional data clustering; finite mixture models; autoregressive dependence; EM algorithm; ERA5-Land; monsoon rainfall

1 Introduction

1.1 Background

Functional data analysis treats each observation as a curve [18, 6]. That is useful when shape and timing matter and a single summary would throw them away. A monsoon season gives one rainfall curve per grid cell and year, and two cells with the same seasonal total can still differ in when the rain arrives. Curves that are observed repeatedly at one location are an example of what Koner and Staicu [12] review as *second-generation functional data*, meaning functional data whose structure goes beyond independent curves.

1.2 Why this matters

Most of Bangladesh's annual rainfall arrives between June and September, and the timing of that rain matters for agriculture and flooding. Grouping locations by how the season unfolds gives a compact description of regional differences. A model that treats thirty or seventy correlated years as independent evidence will be too sure of its clusters and of how many there are.

In my model the cluster label belongs to the unit and does not change across years, while the unit's annual scores follow an AR(1) process. The label describes a lasting spatial grouping, and the autoregressive term describes dependence between years at one location. I will judge the model against versions that ignore the dependence or replace it with a simpler form, and I will not assume the extra structure pays off.

2 Problem Statement, Objectives and Research Questions

If dependence between years is ignored, a mixture can overstate how much information the T curves of a unit carry. That affects uncertainty and, as the pilot suggests, the choice of the number of clusters. An autoregressive term also adds a parameter and may not help in every setting. Simulation can show when it does.

The application is the spatial pattern of Bangladesh monsoon rainfall. It is exploratory. The clusters describe cells with similar modelled trajectories; they are not official meteorological zones or causal climate regimes. The methodological work is the main contribution, and the rainfall data show the method at work. This is not a study of rainfall trends.

The objectives are to:

1. formulate a finite mixture for repeated annual functional scores, with a fixed label per unit and AR(1) dependence across years;
2. derive the observed-data likelihood and implement EM estimation with a numerically optimized autoregressive coefficient;
3. measure clustering accuracy, cluster-number selection, parameter recovery, convergence and run time by simulation, against an independence-based mixture, a random-intercept mixture and, if one exists, a dependence-aware benchmark;
4. fit the model to ERA5-Land monsoon rainfall for Bangladesh, 1950–2025;
5. check how the results change with study period, curve representation, smoothing, score dimension and detrending.

These objectives lead to four research questions.

1. How does AR(1) modelling change clustering accuracy and cluster-number selection relative to an independence-based mixture and to a mixture with persistent unit-level offsets?
2. How well does the procedure recover the clusters and the autoregressive coefficient across sample sizes, dependence strengths and separations?
3. How sensitive are the clusters and the selected number of clusters to imbalance, non-Gaussian innovations, changing variance and a wrong dependence structure?
4. Which spatial groups appear in Bangladesh monsoon rainfall, and how stable are they under reasonable modelling choices?

3 Related Work and Research Gap

Jacques and Preda [9] and Zhang and Parnell [29] review functional clustering, sorting it into basis-score methods, model-based mixtures, distance-based methods and methods for dependent curves. Early model-based approaches work on basis coefficients or principal component scores [11, 5], and mixtures on principal component scores are well established for independent curves [10].

Time series. Xiong and Yeung [27] fit ARMA mixtures by EM; Frühwirth-Schnatter and Kaufmann [7] group several series with finite mixtures; Ren and Barnett [19] cluster by serial correlation; Nguyen et al. [16] add a Markov random field to autoregressive mixtures. These methods work on observed series, not on scores of repeated curves.

Mixed-effects mixtures. Mixtures of linear mixed-effects models give each cluster its own mean trajectory and allow correlated errors within a subject [15, 17]. They share my structure of one label per unit with dependence inside the unit. Because of the overlap, I add a random-intercept mixture as a comparator, so that AR(1) dependence is judged against persistent unit-level offsets as well as against independence.

Repeated curves. Rodriguez and Dunson [20] cluster subjects contributing several curves each using nested Dirichlet process mixtures. As I read the paper, it does not model their order in time.

Dependence among curves. van Delft and Dette [25] build a similarity measure from time-varying spectral density operators; Tang et al. [24] use a functional panel model; López-Oriona et al. [13] use functional quantile autocorrelation; Abramowicz et al. [1] propose a two-stage multiresolution method; Xian [26] fits a Dirichlet process mixture with within-curve correlated errors. These rely on similarity measures, factor structures or within-curve correlation.

Climate applications. Ashkartizabi and Aminghafari [2] cluster climatological curves with K-means and random projection; Saeidi et al. [21] propose Fdmclust, based on a density approximation in a reproducing kernel Hilbert space; Saeidi and Aminghafari [22] propose FunCWM, a cluster-weighted model with functional predictors; Zhang et al. [28] cluster Chinese precipitation stations with a spatial Markov random field after averaging yearly curves, so year-to-year dependence is not modelled.

Table 1 sets these families side by side on four features.

The pieces of my model all exist, but I have not found a published evaluation of their combination for repeated annual curves with a persistent label. The project therefore evaluates a model and does not introduce a new class: its contribution is a worked-out likelihood, a test of when the autoregressive term matters, and a comparison with simpler alternatives. Month 1 will repeat the literature search before any novelty claim is made. Spatial structure and cluster-specific dependence are outside the M.Sc. scope.

4 Methodology

Table 1: Method families compared on four features. The entries are my reading of the cited papers and I will check them again in Month 1.

Method family	Functional curves	Dependence over time	One label per unit	Likelihood on scores
Functional mixtures for independent curves	Yes	No	Yes	Yes
Mixtures of autoregressions for time series	No	Yes	Yes	Yes
Mixed-effects and latent class mixtures	Partly	Within a unit	Yes	Yes
Nested functional clustering	Yes	Not in time order	Yes	No (Bayesian)
Functional time-series clustering	Yes	Yes	Yes	Mostly no
Functional mixture with correlated errors	Yes	Within a curve only	Per curve	Yes
This proposal	Yes	Across years	Yes	Yes

4.1 Data representation

Let $i = 1, \ldots, N$ index units, $t = 1, \ldots, T$ index years, and let s be the position within the season. $Y_{i,t}(s)$ is the curve of unit i in year t, and $z_i \in \{1, \ldots, G\}$ is its fixed cluster label. Pooled functional principal component analysis (FPCA) gives each curve a score vector $\boldsymbol{\xi}_{i,t} \in \mathbb{R}^d$, so all units and years share one basis. I will fix a variance-explained rule for d in advance and vary d in the sensitivity analysis. All models use the same scores.

For Bangladesh, I will build a June–September curve on a common 122-day grid for each cell and year. ERA5-Land daily accumulations can be stamped at the end of the accumulation interval, so I will check timestamps and units against the documentation and sample values before choosing the days.

The main curve is cumulative rainfall through the season. It keeps the timing of accumulation and ends at the seasonal total. A second version divides each curve by its seasonal total, which isolates timing and shape. Seasonal totals are analysed on their own, and I will not interpret normalized curves for zero or near-zero totals. Cumulative curves are monotone, so Gaussian scores are only an approximation. I will check the score distributions and report where they fail.

4.2 Model

Given $z_i = g$, the scores are

$$\boldsymbol{\xi}_{i,t} = \boldsymbol{\mu}_g + u_{i,t}, \qquad u_{i,t} = \phi\, u_{i,t-1} + \boldsymbol{\epsilon}_{i,t}, \qquad |\phi| < 1, \tag{1}$$

with $\boldsymbol{\epsilon}_{i,t} \sim N_d(0, \boldsymbol{\Sigma}_g)$ and $\boldsymbol{\Sigma}_g$ positive definite. The first deviation comes from the stationary distribution, $u_{i,1} \sim N_d\big(0, \boldsymbol{\Sigma}_g/(1-\phi^2)\big)$. The mixing proportions π_g sum to one. One coefficient ϕ serves all clusters and all score components. That keeps the number of dependence parameters small while the number of units is moderate. The model does not describe spatial dependence between cells, and I take this into account when I interpret the clusters.

For a fixed ϕ, define the transformed scores

$$\tilde{\boldsymbol{\xi}}_{i,1} = \sqrt{1-\phi^2}\,\boldsymbol{\xi}_{i,1}, \qquad \tilde{\boldsymbol{\xi}}_{i,t} = \boldsymbol{\xi}_{i,t} - \phi\,\boldsymbol{\xi}_{i,t-1} \quad (t \geq 2). \tag{2}$$

Under cluster g, $\tilde{\boldsymbol{\xi}}_{i,t} \sim N_d(a_t(\phi)\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g)$, independently over t, with $a_1(\phi) = \sqrt{1-\phi^2}$ and $a_t(\phi) = 1-\phi$ for $t \geq 2$. The map from the original to the transformed scores of a unit is lower triangular with determinant $(1-\phi^2)^{d/2}$. The observed-data likelihood is therefore

$$L(\boldsymbol{\Theta}) = \prod_{i=1}^{N} \left[(1-\phi^2)^{d/2} \sum_{g=1}^{G} \pi_g \prod_{t=1}^{T} f_d\Big(\tilde{\boldsymbol{\xi}}_{i,t};\, a_t(\phi)\boldsymbol{\mu}_g,\, \boldsymbol{\Sigma}_g\Big) \right], \tag{3}$$

where f_d is the d-variate normal density and $\boldsymbol{\Theta} = (\pi_g, \boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g, \phi)$.

The Jacobian is the same for every cluster, so it drops out of the cluster responsibilities at a fixed ϕ. It has to stay in when ϕ is optimized. I checked the identity numerically on 45 combinations of $d \in \{1,2,3\}$, $T \in \{4,7,12\}$ and $\phi \in \{-0.8, -0.3, 0, 0.37, 0.8\}$. The transformed density with its Jacobian matched the direct stationary AR(1) density to within 2×10^{-14} in every case.

4.3 Estimation

E-step. The responsibility of cluster g for unit i is

$$\tau_{i,g} = \frac{\pi_g \prod_t f_d\Big(\tilde{\boldsymbol{\xi}}_{i,t};\, a_t(\phi)\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g\Big)}{\sum_h \pi_h \prod_t f_d\big(\tilde{\boldsymbol{\xi}}_{i,t};\, a_t(\phi)\boldsymbol{\mu}_h, \boldsymbol{\Sigma}_h\big)}, \tag{4}$$

computed on the log scale with Cholesky factors and the log-sum-exp trick.

M-step at fixed ϕ.

$$\begin{aligned}
\hat{\pi}_g &= \frac{1}{N}\sum_{i=1}^{N}\tau_{i,g},\\
\hat{\boldsymbol{\mu}}_g &= \frac{\sum_{i=1}^{N}\tau_{i,g}\sum_{t=1}^{T}a_t(\phi)\tilde{\boldsymbol{\xi}}_{i,t}}{\sum_{i=1}^{N}\tau_{i,g}\sum_{t=1}^{T}a_t(\phi)^2},\\
\hat{\boldsymbol{\Sigma}}_g &= \frac{\sum_{i=1}^{N}\tau_{i,g}\sum_{t=1}^{T}r_{i,t,g}\,r_{i,t,g}^{\top}}{T\sum_{i=1}^{N}\tau_{i,g}},
\end{aligned} \tag{5}$$

where $r_{i,t,g} = \tilde{\boldsymbol{\xi}}_{i,t} - a_t(\phi)\hat{\boldsymbol{\mu}}_g$. If a covariance estimate is nearly singular I add a small ridge and count how often that happens. When N is small I will use the parsimonious covariance forms of Celeux and Govaert [4].

Update of ϕ. I maximize the expected complete-data log-likelihood over ϕ with a bounded one-dimensional search. The Jacobian term $\frac{Nd}{2}\log(1-\phi^2)$ stays in the objective, and each candidate ϕ needs freshly transformed scores and means. Because the transformed data depend on ϕ, this is a generalized EM algorithm, and I monitor convergence through the observed-data log-likelihood.

The pilot code has no safeguard against a decrease in that likelihood, and I will add one before the main experiments. After each proposed update, the code recomputes the full observed-data log-likelihood, including the stationary first-year term and the Jacobian. It accepts the update if the likelihood has not dropped by more than a small tolerance. Otherwise it tries a damped update or a direct optimization of the observed-data likelihood, and if neither works it keeps the old parameters and flags the fit. Rejected updates will be logged so that I can report how often they occur. The six diagnostic fits I have run so far showed no meaningful decrease. Six fits do not show that the likelihood never falls.

Initialization. Each fit starts from k-means on unit-level mean scores, random labels and other unit summaries, with initial ϕ values that include zero. I keep the run with the highest converged log-likelihood. Empty clusters, singular covariances and label switching are handled explicitly.

4.4 Choosing the number of clusters

I compare candidate cluster counts with BIC [23] and the integrated completed likelihood (ICL; [3]), and I read both next to cluster sizes, stability across starts and interpretability. In every model considered here, the T scores of a unit share one latent label z_i, and the likelihood factorizes over units, not over the NT curves. The effective sample size is therefore N, the number of independent units, and the BIC penalty uses $\log N$. I will state the parameter count of every fitted model. A reader may ask whether $\log(NT)$ should be used instead, so I will report BIC with $\log(NT)$ as a sensitivity diagnostic. The primary criterion stays $\log N$, because the shared label, not the individual curve, is the independent observation.

BIC and ICL can behave differently under misspecification. Where run time allows, I will therefore also compare the model classes by the held-out log-likelihood of whole score sequences from units left out of the fit. That keeps two questions apart: how many clusters there are, and whether the AR(1) term improves the fit.

4.5 Why ignoring dependence can split clusters

A simple calculation suggests a mechanism for the pilot's over-clustering. Under model (1), the stationary covariance of one deviation is $\Gamma_0 = \boldsymbol{\Sigma}_g/(1-\phi^2)$, and the covariance of the unit's mean score over T years is

$$\operatorname{Var}(\bar{u}_i) = \frac{\Gamma_0}{T}\left[1 + 2\sum_{k=1}^{T-1}\left(1 - \frac{k}{T}\right)\phi^k\right] \approx \frac{\Gamma_0}{T}\cdot\frac{1+\phi}{1-\phi}. \tag{6}$$

A model that treats the T curves as independent expects Γ_0/T. The unit means in fact vary by the factor $(1+\phi)/(1-\phi)$ more than it expects, which is about 1.9 for $\phi = 0.3$ and 4 for $\phi = 0.6$. Put differently, T correlated curves carry about $T_{\text{eff}} = T(1-\phi)/(1+\phi)$ independent curves of information, roughly 11 of 45 when $\phi = 0.6$. A unit whose mean lies a little away from its cluster centre then looks far away to the independence model, and an extra cluster can absorb such units. The gain in log-likelihood grows with T, while the BIC penalty does not.

This is a hypothesis about a mechanism and not a result. I will test it in the simulations by comparing how the over-splitting grows with ϕ and with T, and by checking whether the random-intercept mixture, which also allows unit means to vary, removes it. A first check is reported in Section 5.2.

4.6 Comparison methods

The comparators share the same scores, candidate values of G and number of starts.

1. *Independence mixture.* The same mixture with ϕ fixed at zero.
2. *Independence mixture with a crude effective-sample-size correction.* The independence log-likelihood is multiplied by $(1-\hat{r})/(1+\hat{r})$, with $\hat{r}$ the pooled lag-1 autocorrelation of the scores.
3. *Random-intercept mixture.* Given $z_i = g$, $\xi_{i,t} = \mu_g + b_i + \epsilon_{i,t}$ with $b_i \sim N_d(0, \Omega_g)$ and $\epsilon_{i,t} \sim N_d(0, \Sigma_g)$. The likelihood has a closed form, because the unit mean is distributed as $N_d(\mu_g, \Omega_g + \Sigma_g/T)$ and the deviations from it are independent of the mean. This model describes persistent offsets between units without a time order.
4. *AR(1) mixture.* The proposed model.
5. *Dependence-aware clustering (optional).* If an implementation of a method such as that of López-Oriona et al. [13] can fit repeated curves with fixed unit-level labels, I will add it. Otherwise I will drop it and will not substitute a weaker method.

5 Simulation Study

The true number of clusters is $G = 3$ and fits use $G \in \{2, 3, 4\}$. The main design takes $N \in \{30, 60\}$ and $T \in \{45, 75\}$, plus a stress test with $N = 15$, $T = 45$, which gives five (N, T) combinations. The primary generator creates serial dependence directly at the score level, which is the level at which the models are fitted. For unit i in cluster g,

$$\xi_{i,t} - \boldsymbol{\mu}_g = \phi(\xi_{i,t-1} - \boldsymbol{\mu}_g) + \boldsymbol{\epsilon}_{i,t}, \qquad \boldsymbol{\epsilon}_{i,t} \sim N_d(0, \boldsymbol{\Sigma}_g), \tag{7}$$

and the first deviation is drawn from its stationary distribution. The cluster means $\boldsymbol{\mu}_g$ are chosen to give the target separation. A smaller supplementary check, run for the baseline only, builds curves from these scores on a common basis, adds Matérn ($\nu = 3/2$) noise, and applies pooled FPCA before fitting. It shows what the two-stage step costs. Matérn noise alone does not create year-to-year dependence, so it is never used as a stand-in for it.

The baseline has $\phi = 0.3$, Gaussian innovations, balanced clusters and good separation. The other nine scenarios each change one feature:

- *Dependence:* $\phi = 0$ and $\phi = 0.6$.

- *Changing variance:* the innovation variance is multiplied by 2.5 after the midpoint; this is a deliberate misspecification, because the fitted covariance stays fixed.
- *Heavy tails:* Student-t innovations with 3 degrees of freedom, rescaled to unit variance.
- *Weak separation:* cluster means about 0.4 innovation standard deviations apart.
- *Imbalance:* mixing proportions 0.5, 0.3 and 0.2.
- *Cluster-specific dependence:* $\phi = 0.3$ in one cluster and $\phi = 0.8$ in another, which breaks the common-ϕ assumption.
- *Higher-order dependence:* deviations generated by an AR(2) process with a similar first-lag correlation.
- *Persistent unit offsets:* no autoregression, but a random intercept b_i with standard deviation about half the innovation standard deviation; this is where the random-intercept mixture is the correct model.

This gives ten scenarios and 50 design cells. With 500 replications per cell the study generates 25,000 datasets. Fitting the independence and AR(1) mixtures for three values of G gives 150,000 fits before multiple starts; the random-intercept mixture, fitted in the baseline, dependence and persistent-offset scenarios, adds about 30,000 more. I will first benchmark run time on a representative subset. If the full design does not fit the schedule, I will run a screening pass with fewer replications in every cell, then 500 replications for the main scenarios and for cells where the models behave unclearly. Any reduction will be reported, and every scenario stays in the design. A spatially correlated version of the baseline is a possible addition if time allows, and I will also test run time and stability on a run with several hundred units, because the Bangladesh grid has on the order of a thousand cells.

For each setting I report the adjusted Rand index (ARI; [8]), the share of replications in which BIC and ICL pick $G = 3$, the bias and RMSE of $\hat{\phi}$, the error of the estimated cluster means after aligning labels, the convergence rate, the number of iterations, the number of ridge adjustments and the run time. All come with Monte Carlo standard errors. I do not expect AR(1) modelling to win everywhere, and the results will show where it helps and where it does not.

5.1 Pilot study

To test the code and the main idea, I ran a small pilot in Python. It draws three equal clusters of $d = 2$ scores with identity innovation covariance, Gaussian innovations and cluster means

about 1.1 innovation standard deviations apart. Each replication fits the independence and AR(1) mixtures for $G = 2, 3, 4$ from three k-means starts and picks G by BIC. I ran 60 replications at $N = 30$, $T = 45$ and 40 at $N = 60$, $T = 75$. The code and seeds are saved. A rerun of the $N = 30$, $T = 45$ configuration with the saved seeds reproduced all 360 stored comparisons. I have not yet repeated that check for $N = 60$, $T = 75$, and a full row-by-row check of Table 2 against the stored output is planned for Month 2.

Table 2: Pilot results. ARI is for the $G = 3$ fit. Percentages show how often BIC chose each G. Choices not listed were 0%, except that the AR(1) mixture chose $G = 4$ in 2% of replications at $N = 30$, $T = 45$, $\phi = 0$.

Setting	ϕ	Independence mixture			AR(1) mixture			Mean $\hat{\phi}$
		ARI	G=3 (%)	G=4 (%)	ARI	G=3 (%)	G=2 (%)	
$N = 30,\ T = 45$	0.0	1.000	98	2	1.000	98	0	−0.004
$N = 30,\ T = 45$	0.3	0.974	67	33	0.976	100	0	0.298
$N = 30,\ T = 45$	0.6	0.619	2	98	0.559	28	72	0.595
$N = 60,\ T = 75$	0.0	1.000	100	0	1.000	100	0	−0.002
$N = 60,\ T = 75$	0.3	0.992	75	25	0.994	100	0	0.296
$N = 60,\ T = 75$	0.6	0.843	0	100	0.831	97.5	2.5	0.599

At $\phi = 0$ the two models behave alike, as they should. At $\phi = 0.3$ the independence mixture chose four clusters in a quarter to a third of the replications, and at $\phi = 0.6$ in 98% and 100%. The AR(1) mixture chose three clusters in every replication at $\phi = 0.3$ and in 39 of 40 replications at $\phi = 0.6$ with $N = 60$, $T = 75$. The ARI barely moved between the two models in most rows. The exception is $N = 30$, $T = 45$, $\phi = 0.6$, where the independence mixture scored higher (0.619 against 0.559) and the AR(1) mixture chose two clusters in 72% of replications. With strong persistence and few years, the unit means are noisy and the clusters are hard to tell apart.

Each cell rests on 40 or 60 replications, so a percentage carries a standard error of up to about 8 points, and with 40 replications a single run moves a percentage by 2.5 points. The full study will report counts next to percentages. Reruns with other seeds moved some cells by more than that. The data also come from one design that favours the AR(1) model, since they follow its assumptions exactly.

5.2 Additional checks and the random-intercept comparison

A small new check asks what the random-intercept mixture does at $N = 30$, $T = 45$, $\phi = 0.6$, the cell where the earlier pilot had been least stable. In this check the AR(1) mixture chose two clusters in 27 of 60 replications, against 72% in Table 2, so that cell is indeed unstable.

The check is small, uses data generated from the AR(1) model, and does not yet test the random-intercept alternative on its own ground. It supports running the full comparison in Section 4.6, where the random-intercept mixture is fitted alongside the independence and AR(1) mixtures, including in the persistent-offset scenario.

6 Application: Bangladesh Monsoon Rainfall

The data are ERA5-Land reanalysis precipitation [14] for Bangladesh, 1950–2025, subject to quality checks. I will define the domain with a documented national boundary and a cell inclusion rule. Cells with incomplete or unusable records will be identified before the analysis, and I will keep those that meet a completeness criterion fixed in advance and report how many cells were dropped and where. I will check units, accumulation intervals, time coordinates, missing values and duplicate dates, and convert to millimetres using the convention I verify. Reanalysis quality can change over time, so I will repeat the main analysis for 1981–2025. That is a check, and it does not mean the early years are unusable.

A first quality check of the 76 annual files found complete 1 June–30 September coverage in every year, no duplicate dates and no negative values. The check used a rectangular box and not the national boundary, so it included cells in neighbouring countries, and the sea cells were missing in every year, as expected. Quality summaries will also cover seasonal totals, zero or near-zero totals, implausible values and breaks introduced by extraction. Smoothing will only remove high-frequency noise, and I will compare raw and smoothed curves.

Before fitting any mixture I will compute the lag-1 autocorrelation of the pooled scores. The proposal rests on the idea that annual curves are correlated. If the real data show little persistence, $\hat{\phi}$ will sit near zero and the models will agree, and I will report that as the result.

I compare candidate numbers of clusters from $G = 1$ to $G = 8$ by BIC and ICL, cluster size, stability and interpretability. As a simple reference, I also cluster the cells on seasonal total rainfall alone and report how much the trajectory-based groups differ from it. This separates groups driven by rainfall amount from groups driven by timing. The days on which cumulative rainfall reaches 10%, 50% and 90% of the season may help describe the clusters; they are not official onset or withdrawal dates, and I will not call them that. Maps will show the assigned clusters, and cluster-average curves will show typical seasonal patterns.

I assess uncertainty through repeated starts and through spatial block resampling: I withhold contiguous blocks of cells, refit, and check whether the remaining clusters keep

their shape. To compare the full-data and block-held-out partitions I align cluster labels by the Hungarian algorithm on the confusion matrix and report the adjusted Rand index between the two partitions as a stability measure, together with the fraction of cells whose aligned label is unchanged. The sensitivity checks vary the period (1950–2025 against 1981–2025), the curve (cumulative against normalized), the smoothing level, the score dimension, the cell inclusion rule and detrending of the scores. A fixed cluster mean with AR(1) errors can absorb a long-term trend as apparent persistence, so detrending shows whether $\hat{\phi}$ and the clusters depend on a trend. Detrending can also remove a real climate signal, and I will say so. Neighbouring cells are not independent, so I will not treat them as independent replications of the climate.

7 Expected Outcomes and Limitations

I expect to deliver:

1. a mixture model for repeated annual curves with a fixed label per unit and AR(1) dependence across years, together with its likelihood;
2. a tested EM implementation with the likelihood safeguard described in Section 4.3;
3. a reproducible simulation study showing when modelling dependence helps, when it makes no difference and when it hurts, including a comparison with a random-intercept mixture;
4. an exploratory grouping of Bangladesh grid cells by monsoon trajectory, with maps, cluster-average curves and sensitivity checks;
5. analysis code and a record of preprocessing choices and diagnostics.

The model leaves out spatial dependence, abrupt regime changes within a unit and fully nonstationary dependence. Other limits are:

- AR(1) may not capture all persistence in annual rainfall, and one ϕ for all clusters and score components may be too restrictive. Allowing a coefficient per component is a natural extension, but it changes the stationary start and I leave it for later work.
- FPCA is run once on the pooled curves before clustering, so uncertainty in the scores is not carried into the mixture.
- Gaussian innovations may react strongly to extreme years; the heavy-tail scenario tests this.

- Clusters of grid cells may reflect spatial correlation and shared large-scale forcing that the model cannot separate.
- Reanalysis is not direct observation and may carry bias that changes over time.
- Results may depend on normalization, smoothing, the score dimension and the chosen G.

8 Work Plan

Table 3: Proposed six-month schedule, counted from the start of the project. Data preparation and ERA5-Land quality control run in parallel with the simulations, so the two strands finish in the same month.

Month	Work
1	Literature search and updated novelty check; notation; likelihood derivation and identifiability checks; ERA5-Land download, timestamp and unit checks; freeze preprocessing rules and seeds
2	Implement the independence, random-intercept and AR(1) mixtures on a shared skeleton; add the likelihood safeguard; validate on small simulated examples; time the pilot code; fix the score dimension and BIC rules
3	Simulation screening pass with reduced replications; record per-fit run time; begin the main replication run; start ERA5-Land quality control and build the annual curves
4	Complete the main simulation replications and summaries of accuracy, parameter recovery and run time; fit all models to the Bangladesh curves; begin sensitivity analyses
5	Finish ERA5-Land sensitivity analyses (period, curve, smoothing, score dimension, cell inclusion, detrending); stability checks and maps; assemble simulation and application figures and tables
6	Interpretation; thesis writing; revision with the supervisor; code and data documentation release

The plan is tight. It assumes that the three models can share one implementation skeleton and that the screening pass does not expose a numerical problem. The two main risks are simulation run time and the ERA5-Land sensitivity grid. If either overruns, I will reduce the replication count in the secondary scenarios and report the reduction. I will not drop a scenario.

9 Data Management and Ethics

The project uses secondary gridded climate data and involves no people or personal data. I will record data access dates, preprocessing choices, software versions, random seeds, model settings and convergence rules, and I will follow the data provider's terms for any redistribution. Maps will state that the clusters are model-based groupings and not official climate classifications.

References

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